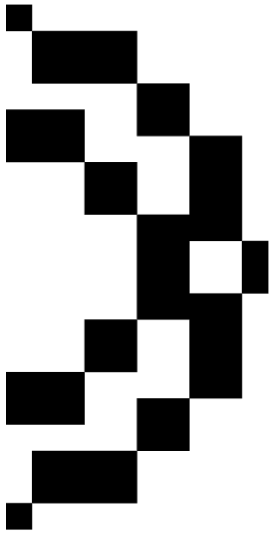
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Course Guide

Computation I · 2026/27

Prof. Karina Brotto Rebuli
krebuli@novaims.unl.pt

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01.

Why you're here

The skill that didn't get automated

In December, you will be able to

What this course is actually for

Read a program written by someone — or something — else, and say what it does.

Find the line where it goes wrong, and explain why it goes wrong there.

Write correct programs from scratch, in Python, without help.

Judge whether a solution is good, not merely whether it runs.

Prepare you for Computation II.

Why this matters now

The honest version

Generating code that runs is no longer difficult. You can already do it, today, without this course.

But a machine checks only whether your instructions can be executed — never whether they mean what you intended.

Finding where the two diverge is still a human's job. So is standing behind the result: code you ship is code you answer for, with or without AI.

That is the expensive work, and it is the work nobody has automated.

Fourteen weeks from now, you will be the person in the room who can do both.

02.

How the course runs

Three classes a week, two different jobs

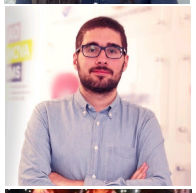
The team

Who to ask, and about what



Karina Brotto Rebuli — lectures, course design, anything about the theory.

krebuli@novaims.unl.pt



Diogo Rasteiro — practical class, groups P4, P5, and PISPA.

drasteiro@novaims.unl.pt



Liah Rosenfeld — practical class, groups P1, P2, and P3.

lrosenfeld@novaims.unl.pt



Samuel Santos — practical class, groups P6 and P7.

samuel.santos@novaims.unl.pt

Not sure which Practical Group is yours? Ask Academic Services (this week, not in week three).

Your week

Theory and practice do different things — you need both

Lecture · 1h30 Concepts, live code, and the reasoning behind them. Two groups.

Practical classes · 3h per week Two sessions per week for each Practical Group (they start in week 2).

The book · A weekly reading map tells you exactly what to read.

Moodle quizzes · At home, for self-assessment. Announced on Moodle as each one opens.

After each lecture ⟶ Read the Study PDF (your copy of the lecture, with study notes), do the week's exercise pack, and the quiz, all on Moodle.

Take notes during the classes!! Your notes should become your first reference.

Your first two weeks

They run a little differently — here is the map

Week 1 · Lectures only — no practical classes yet. Today and one extra lecture, Thursday 10/9 or Friday 11/9: you vote today, and the day is on Moodle this afternoon.

Monday 14/9 · Your first practical class: installing Python and PyCharm, in this room from **11h to 13h**, with Prof. Diogo Rasteiro. Bring your laptop.

Week 2 onwards · practical classes run normally, two sessions a week for all Practical Groups.

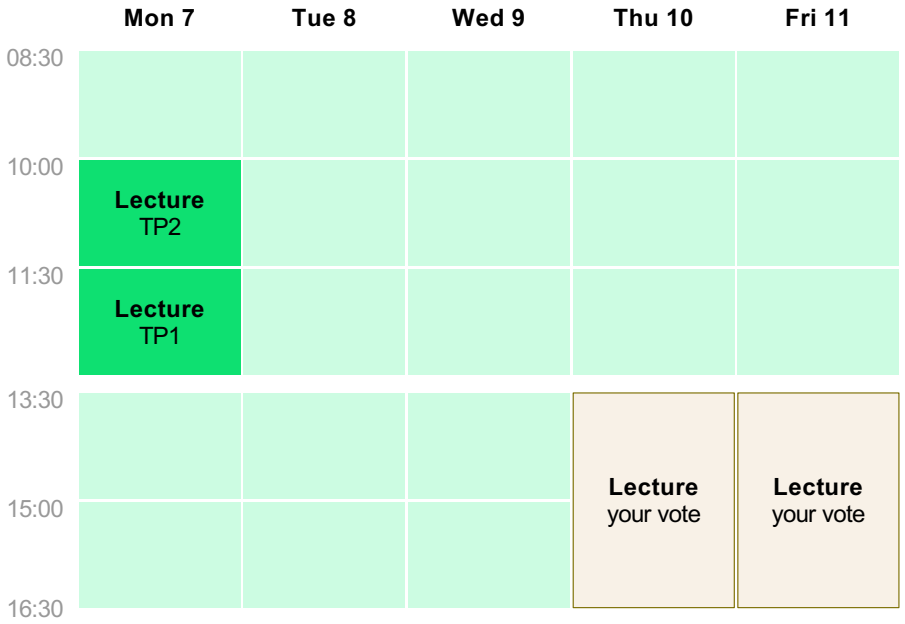
The week-1 practical sessions are not lost — they come back as extra classes before the coding tests.

The first two weeks of Computation I, day by day

Computation I only — the rest of your timetable stays as it is

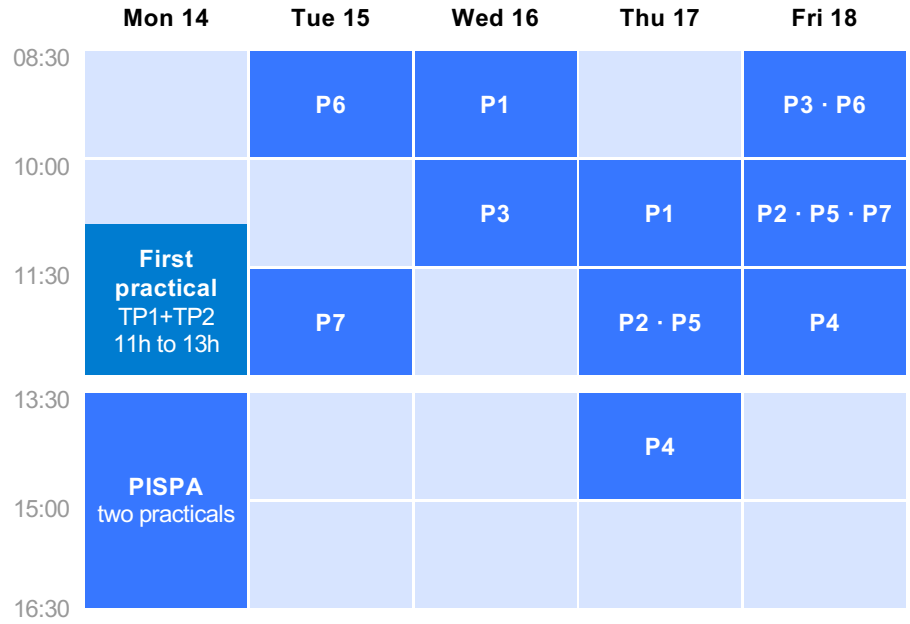
Week 1 · 7–11 September

Only theoretical lectures



Week 2 · 14–18 September

Only practical classes



03.

The book

One book, read properly, is enough

Think Python, 3rd edition

Allen B. Downey — free, legally, in HTML and PDF

The book is free under a Creative Commons licence. There is nothing to buy:

https://allendowney.github.io/ThinkPython/

Our lectures do not follow the book's order. A weekly reading map on Moodle tells you which sections to read, which to skip, and which to postpone.

Read the Debugging section at the end of every chapter. It is not optional.

Prefer Portuguese, or want to go deeper? *Programação em Python — Fundamentos e Resolução de Problemas* (E. Costa) is in the NOVA IMS library.

04.

Assessment

Three moments, and none of them is a surprise

What counts, and how much

The final grade is one line of Python

```
final = exam * 0.7 + ((coding_test1 + coding_test2) / 2) * 0.3 - behaviour_penalty
```

Final exam · 70% Your grade in the first or second epoch (see *Coding tests and the exam* and *The second epoch*). On paper, a set of exercises and questions that cover all the program of the discipline.

Two coding tests · 30% The average of the two coding tests. Written, on paper, Saturdays of weeks 7 and 13.

`Behaviour penalty` · Normally zero. It exists so that it never has to be used.

Moodle quizzes · 0% Not graded. Which is exactly why you should do them.

Coding tests and the exam

The days on paper — same rules

Miss one, and that evaluation is a zero. **There are no alternative dates, for any reason.**

No notes, no books, no documentation of any kind. Calculators only if non-graphical.

Phones and smart devices switched off and **left at the entrance of the room**. Touching one during an evaluation counts as cheating — whatever it was for.

Cheating, in any form, fails the discipline for the whole academic year: zero, in all epochs.

This is a NOVA IMS rule.

The fine print

Four details worth knowing before you need them

All grades live on the Portuguese 0–20 scale. **Ten passes**.

Partial grades enter the formula with all their decimals. **Only the final result is rounded**, to the nearest integer: e.g. 14.499 is a 14; 14.5 is a 15.

Typical cases of indiscipline that lead to a behaviour penalty include, but are not limited to, the following:

- Making noise during (theoretical or practical) classes.
- Consulting material (on computers, portable devices, on paper, etc.) that does not concern the course of Computation I during a (theoretical or practical) class.
- Using the mobile phone during the (theoretical or practical) classes.
- Encouraging or favouring cheating in any way; **etc…**

⟶ All cases of indiscipline, decided by the unquestionable opinion of the teachers, besides causing a penalty in the final evaluations, will also be reported to the Director of the course and may have disciplinary consequences.

Absence in evaluations

There is absolutely no way that a student can do a test or exam on a different day

If a student is absent from one or more evaluations, be it a coding test or the exam, their grade for that specific evaluation **will be equal to zero.**

You can go to the final exam also if you have zero in the coding test evaluations.

You can pass even in case the grade of the final exam is < 10, as long as the global evaluation is ≥ 10.

Attention: you cannot be absent from the final exam. Presence in the final exam is, of course, mandatory.

The second epoch

One more exam, on harder terms

Portugal gives you two examination epochs per year. **The second epoch is a single exam that replaces your first-epoch exam grade.** The two are never averaged, and only the higher of them counts.

If you failed the first epoch, you sit it for free; if you are improving a pass, there is a small fee.

It is also the harder exam. If you can pass in the first epoch, do. The behaviour penalty follows you there too.

Coding tests do not have a second epoch — they cannot be retaken, for any reason. The 30% you earned in October and December is the 30% you carry into either epoch.

Details and dates: Academic Services.

Plagiarism · Copy · Cheating

Not acceptable, this is the rule of NOVA IMS

If a student is caught copying or cheating in any possible way, during the exam or the coding test, they will fail the discipline for the whole academic year (receiving a grade equal to zero in all the examination epochs of this year, both before and after the cheating episode).

Using, touching or watching a mobile phone or smartwatch during a coding test or an exam will always be considered a case of plagiarism/copy/cheating regardless of what it was used for.

All cases of plagiarism/copy/cheating are decided by the **unquestionable** opinion of the teachers.

The Moodle quizzes

From easy to challenging, and worth no marks — both on purpose.

Quizzes should give you a better picture of your own knowledge of the discipline.

They open on Moodle throughout the semester. You'll be told when each one opens — and, with it, the number of attempts and the deadline.

Most questions ask you to read a piece of code and improve it, rather than write one from scratch.

The questions range from easy to hard, so you can assess how much you've learned.

Full answers are released on Moodle afterwards.

05.

The fourteen weeks

Where we're going, and when

Our fourteen weeks

First half · weeks 1 to 7

Week	Date	Lecture	Extra
1	7/9	What a program is. The interpreter, and the three kinds of error	Lecture 10 or 11/9 — replaces Mon 14/9
2	10 or 11/9	Variables, data types, assignment — and the arithmetic and relational operators	
3	21/9	Logical expressions, and strings as sequences	
4	28/9	Input and output, the Python console — and lists	Lecture replacing Mon 5/10 — voted later
5	TBD	The conditional instruction: if	
6	12/10	The iterative instruction: for, and range	
7	19/10	Lists and for, together	Coding test 1 (Saturday · 24/10)

Our fourteen weeks

Second half · weeks 8 to 14

Week	Date	Lecture	Extra
8	**26/10**	The iterative instruction: while, and break	
9	**2/11**	What went wrong in coding test 1 — and nested for	
10	**9/11**	List comprehensions	
11	**16/11**	Matrices	
12	**23/11**	Functions: definition, call, return	
13	**30/11**	Functions: parameters, scope	**Coding test 2 (Saturday · 5/12)**
14	**7/12**	Modules and other Python components. Where you go next	

Choose the day for replacing the 14/9 lecture

The lecture of Monday 14/9 needs a new day — pick it

Thursday
10 September
afternoon

Friday
11 September
afternoon

both days in the 13:30–16:30 window

The exact day and time for each theoretical group (TP) will be published on Moodle, today, by the end of the afternoon.

The lecture of Monday 5/10 gets its own vote, later in the course.

06.

Working rules

Short list, seriously meant

Using AI in this course

Learning to use AI without unlearning how to think

You may use AI tools for your own study. We are not going to pretend otherwise.

You may not present as your own any work you cannot explain. The main question you have to ask yourself is: Can you defend every line of it to someone who asks?

The coding tests and the final exam are **on paper**, with no devices. This is not a trap — it has always been so, and it is the only honest measure of what you personally can do.

Getting started

Do it before the first practical

A laptop — you will need it in every theoretical and practical class. Don't have one? IT Services can lend you one — ask before your first practical.

Python 3 — free. Installation guide and video on Moodle.

PyCharm Community Edition — installation guide and video on Moodle.

No luck? — you can run Python in a browser for week 1. Details in the setup guide.

Stuck? — bring the laptop to the first practical (Mon 14/9). That is what it is for.

Where to get help

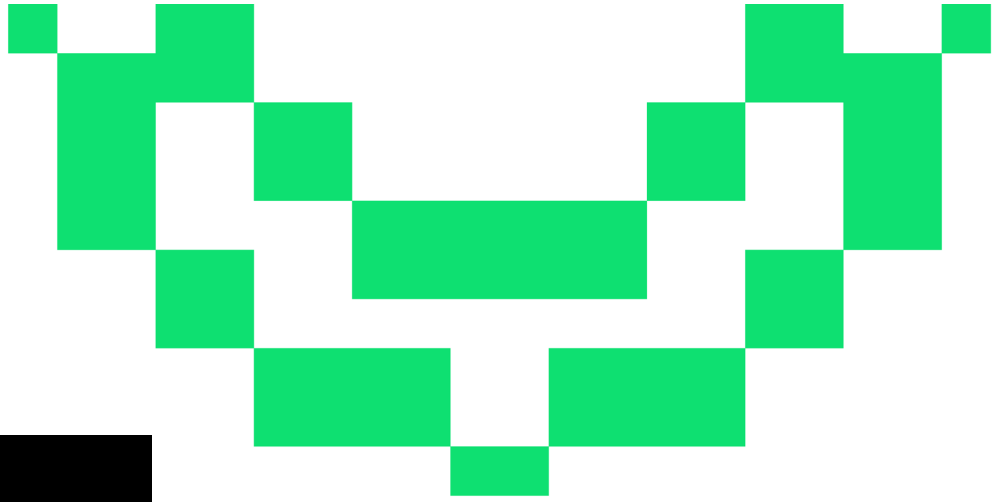
In roughly this order

The practical class. It is staffed, and it is the cheapest help you will ever get.

The Moodle forum — ask in public. Someone else has the same question.

The Debugging section of the chapter you are on. Genuinely.

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```
input("Questions? ")
```

Acreditações e Certificações da NOVA IMS

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